long, which can pack four times more data carriers using lower sub-carrier spacing.

Symbol length and sub-carrier frequency are interlinked, which means if frequency reduces, symbol length increases.

The advantage of having a longer symbol length is that one is more immune to delays of signal reception. So, in case of multiple signal reflections, each reflection is slightly delayed. When that happens, the longer the symbol length, the more robust it is, without inter-symbol interference.

Longer symbol length helps achieve a bigger guard interval, which is 3.2 µs versus 0.8 µs. A bigger guard interval safeguards from multiple reflections and helps get more reliable data. This improves average throughput in dense environments when connections to AP increases.

Apart from these two major changes, speed has increased, which is the main purpose of WLAN. 256 QAM has gone up to 1024 QAM, which is almost a 25 per cent increase in data rate.

Data rates of 600Mbps can be obtained for a single spatial stream and 80MHz BW. WLAN6 allows for eight spatial uplink and downlink streams making multiple gigabits/sec data rates possible.

Sub-carrier spacing reduction enables packing more sub-carriers in 20MHz channel bandwidth

Wi-Fi 5 has 52 sub-carriers in 20MHz bandwidth. Whereas, Wi-Fi 6 packs 242 sub-carriers in the same 20MHz channel. These sub-carriers can be divided into nine parts called resource units. Minimum size for a resource unit is 26 sub-carriers and 2MHz bandwidth.

Wi-Fi 5 allocates minimum 20MHz bandwidth to a single user irrespective of the quantity of data packet. So, data-intensive video downloads and low-data activities like sending a text message both get the same allocation of bandwidth. This is a waste of precious spectrum resource.

With Wi-Fi 6, spectrum space can be more efficiently utilised with OFDMA technology, which allocates lower bandwidth (2MHz) to low-data-rate activities and high bandwidth to higher-data-rate applications. It also allows multiple low-data-rate activities from different users/devices to be served in the same channel at the same time, improving system efficiency.

11ax features that enable the IoT are as follows:

- OFDMA for both uplink as well as downlink. It offers more spectrum efficiency and more network capacity. More users can be served at one time.
- 8 × 8 multi-user MIMO uplink and downlink. It allows transmitting to eight users together for both transmission and reception.
- Long OFDM symbol duration.
- Power save modes with longer sleep intervals and scheduled wake-up times.

In terms of future IoT use-cases that 11ax enables, there are battery-operated devices for video surveillance, remote diagnostics, disaster management and so on. All these use-cases require low power, long range, low latency as well as good data rate.

There are augmented and virtual reality (AR and VR) kind of applications that Wi-Fi 6 enables in high data rate modes. Virtual training, virtual diagnostics, robotic surgery, AR tourism and others are some of the new applications that have become possible with Wi-Fi 6.

Bluetooth 5: the latest technology from Bluetooth

The previous generation, Bluetooth 4, had two sub-parts: classic and low energy. Classic was mostly used in headsets and sound buds. Low energy is used in wearables, fitness bands and similar applications.

Bluetooth 5 has new features that enable the IoT. Some of the features are given below.

- It gives 4x longer range for low data rates. This is enabled by adding more forward error correction to the data that is sent. For every bit of data that is sent, this mode sends two more error correction codes that improve the chances of detecting that bit, giving four times the range. But this is only for low data rates.
- Bluetooth 5 can achieve 2Mbps data rate. Faster data rate is achieved by doubling symbol rate. This helps with faster pairing, connection and download speed.

Earlier, Bluetooth 4 had 31 bytes for advertising, which is basically broadcast messages. This has now been increased eight times, thus helping achieve richer advertising content.

Various applications of Bluetooth 5 include health and fitness (heart rate monitors, smart bands), wireless audio (soundbars, speakers, headsets), retail (location-based ads) and so on. Most of these are like Bluetooth 4, but their specifications have improved, with lower power and better range.

Wireless standards that are specifically meant for the IoT

Sub-1GHz is specifically meant for IoT applications. It gives very long range (as band is from 750MHz to 950MHz), possibly up to 20km distance. It can run for multiple years on a single coin battery. This robust technology uses frequency hopping and is less susceptible to interference, and avoids the crowded 2.4GHz space.

Its applications include streetlight controls, e-meters, smoke detectors, toll road tax systems, irrigation systems and the like.

NB-IoT is the latest LTE standard. It is the lowest data rate LTE version. In LTE, there is the high-performance mode that offers up to 1Gbit/s kind of data rates. This is where 5G is going to come. At the other end of the spectrum, it has low cost, power and throughput. The range is 20km with data rate of 250Kbps. Use-cases are similar to sub-1GHz.

Choosing the right wireless standard

Depending on the use-case, the communication technology should be selected. Except for NB-IoT, which is subscription-based, all others are licence-free. While NB-IoT and sub-1GHz offer a good range, Wi-Fi is the best choice for speed. Bluetooth, sub-1GHz and NB-IoT give similar battery life.

As per a Digi-Key report, cost is the highest for NB-IoT at ₹ 1000. Cost for both Wi-Fi and Bluetooth is ₹ 300, while that of sub-1GHz is higher at ₹ 350. These are module costs, including antenna. EFY